

TrES-3b

R-1988

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_130525 · created 2026-08-07T14:51:29+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456437.778 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.188 +/- 0.053
Transit depth (Rp/Rs)^2	3.52 +/- 1.99 [%]
Orbital Inclination (inc)	80.46 +/- 2.55
Airmass coefficient 1 (a1)	1.016 +/- 0.02
Airmass coefficient 2 (a2)	-0.007 +/- 0.022
Scatter in the residuals of the lightcurve fit is	1.96 %
Best Comparison Star	#1 - [396, 349]
Optimal Method	PSF photometry
Transit Duration (day)	0.037 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.188 ± 0.053 vs 0.16309 (deviation 0.43σ)
Duration fit vs published	0.037 d vs 0.0567 d (deviation 0.66σ)
Mid-time O-C	1.8 min
Depth SNR	3.2
Residual scatter	1.96 %

Light curve

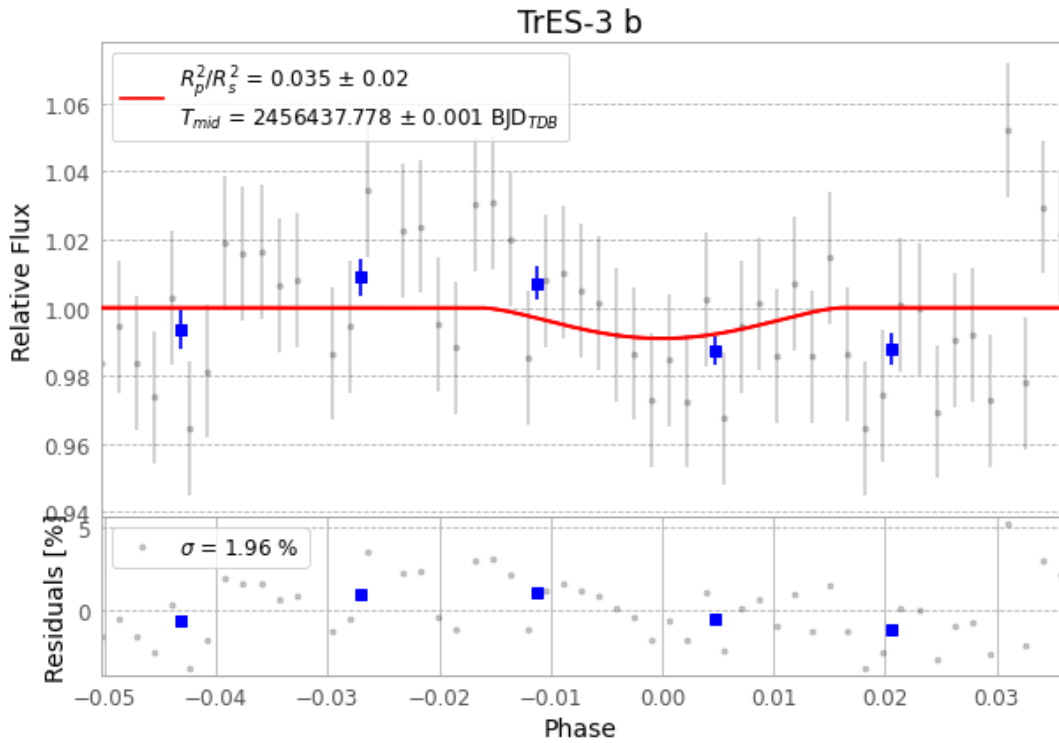


Figure 1 — FinalLightCurve_TrES-3 b_2013-05-24.png (SHA-256 f031908d19f6ddec...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [290, 194], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 b31bc91f29009dfe...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

55 FITS frames (36 MB), TRES-3130525050012.FITS → TRES-3130525074212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-130525.log (14a3f8a16e68...), FinalLightCurve_TrES-3 b_2013-05-24.png (f031908d19f6...), FinalLightCurve_TrES-3 b_2013-05-24.csv (4169db9e1ed4...)

Compute resources

Wall time 120.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 14:51Z.